Geometry Review: Unit 4 — Similarity

Answer Key

Station 1: Ratios and Proportions

1. $x = 3$
2. $x = 8$
3. $x = 9$
4. $x = 6$
5. $\frac{3}{4}$
6. $40°,\ 60°,\ 80°$
7. 35 and 49
8. Yes; $6 \cdot 15 = 9 \cdot 10 = 90$
9. 5 cups
10. 87.5 miles

Station 2: Similar Polygons and Scale Factor

1. $EF = 12$
2. $\frac{5}{2}$
3. $x = 15$
4. 36
5. 50°
6. 35
7. No; $\frac{3}{6} \neq \frac{5}{9}$
8. Yes; scale factor 3
9. $9 : 25$
10. 180 $\quad 20 \cdot 3^2$

Station 3: Proving Triangles Similar: AA, SAS, SSS

1. Yes (AA); the third angles are 70° and 50°
2. Yes; SSS∼, scale factor $\frac{3}{2}$
3. Yes; SAS∼ $(\frac{6}{4} = \frac{9}{6} = \frac{3}{2})$
4. No; $\frac{5}{10} = \frac{7}{14} = \frac{1}{2}$ but $\frac{9}{20} \neq \frac{1}{2}$
5. Yes (AA)
6. AA
7. $\frac{AC}{DF} = \frac{3}{2}$
8. $x = 6$
9. Yes; SSS∼, $\frac{3}{2}$
10. 6 $\quad h^2 = 4 \cdot 9$

Station 4: Applications of Similarity

1. 27 ft
2. 40 ft
3. 40 m
4. $EC = 9$
5. $AC = 20$
6. Yes; $\frac{3}{6} = \frac{4}{8} = \frac{1}{2}$
7. $x = 12$
8. 30 m
9. 49
10. 14 $\quad$ *side ratio* $= \sqrt{25} : \sqrt{49} = 5 : 7$